

Step 1

- Identify states and tape symbols with numbers:

$$Q = \{0, 1, \dots, n\} \quad \Sigma = \{0, 1, \dots, m\}$$

where $s = 0$ and $\sqcup = 0$

- Code configurations $c = (q, w, u)$ with three numbers:

- q , the state number

- $\lceil [a_i, \dots, a_1] \rceil$ where $w = a_1 \cdots a_i$

- $\lceil [b_1, \dots, b_j] \rceil$ where $u = b_1 \cdots b_j$

[The reversal of w makes it easier to use our RM programs for list manipulation.]

- Identify directions with numbers: $L = 0, R = 1$